

A Verification Pipeline for the 5D Flavor Constraint Catalog

5D-Neutrino-Mixing catalog v0.4 (2026-05-17)

What the catalog is. A structured inventory of **102 flavor-physics constraints** on warped extra-dimensional (RS) parameter space: 80 low-energy entries (kaon, charm, beauty, top/Higgs/EW, charged-lepton LFV, EDM/neutrino), 14 direct-collider RS resonance entries (KK-gluon, KK gauge, KK-graviton, custodial top partners, VLQ, EW-precision tail), and 8 deferred-scope re-promotions. Each entry records the measured experimental value or limit with year, uncertainty, source URL, access date, and `sha256` of a tracked local text snapshot. Tags `v0`, `v0.1`, `v0.2`, `v0.3`, `v0.4` pin reproducible states.

How processes were found. Five independent mechanisms, layered. (i) An overcomplete 128-entry seed list assembled from PDG, HFLAV/FLAG, and the core RS-flavor literature (Agashe–Contino–Pomarol, Csaki–Falkowski–Weiler, Perez–Randall). (ii) 10 PI-seed processes (Wave-1, physics judgment). (iii) Four non-overlapping Discovery Agent rounds (DA-1..DA-4), each surveying a distinct slice of the literature against the running inventory. (iv) An *independent external reconciliation*: GPT Deep Research was handed the v0.1 catalog and the broader literature and surfaced 5 entries DA-4 had ambiguously closed (Wave-7: T003, T004, T008, T012, B012). (v) Scope extensions: Wave-8 promoted 8 SECONDARY entries from the DA-deferred tail; Wave-9 opened a new `collider_rs` family of 14 PRIMARY entries under explicit PI directive. The convergence rule (two consecutive DA rounds finding ≤ 1 entry each) was met; 43 remaining plan rows carry DEFERRED-SCOPE tags with named promotion triggers—no silent drops.

How each entry is verified. Every numerical claim survives a five-stage chain before approval. **(1) Process Knowledge Agent** retrieves the canonical PDG / HFLAV / collaboration value and writes a local text snapshot with `sha256`. **(2) Writer Agent** polishes prose and structured metadata. **(3) Checker Agent**—in an *independent* session, never the WA on the same batch—runs an 8-item checklist: every numerical claim traces to a structured `pdg_or_equivalent` block with year, uncertainty, URL, access date, `sha256`; every reference key resolves to a manifest entry; `status_history` transitions legal; in-code coverage claims `grep-match` cited evidence; L^AT_EX hygiene clean. **(4) Fact-Check Agent**, also independent and equipped with WebFetch, *re-fetches every cited URL*, verifies authors/title/year against the manifest, and verifies the specific numerical value appears in both the fetched page and the local snapshot. **(5) Opus sign-off** reviews the chain and emits APPROVE / RETURN-TO-WA / ESCALATE-TO-PI, with a per-round verdict index committed to git. Writer/checker separation is non-negotiable; in a sister artifact two BLOCKER-class physics errors were caught only because the CA was genuinely independent.

Two concrete catches. *T020* ($h \rightarrow e\mu$). The catalog recorded $\mathcal{B}(h \rightarrow e\mu) < 6.1 \times 10^{-5}$. The fact-check WebFetch hit the live [ATLAS source](#) and saw $< 6.2 \times 10^{-5}$ observed ($< 5.9 \times 10^{-5}$ expected). A one-digit slip, fixed and re-fact-checked. It is the **only** numerical mismatch caught across all 102 entries—exactly the class of error single-agent self-review cannot find. *K020* ($K^+ \rightarrow \pi^+ e^\pm \mu^\mp$). The Checker Agent flagged that a measured upper limit ($< 2.1 \times 10^{-11}$ at 90% CL, Sher / BNL E865) appeared in prose but was missing from the structured `pdg_or_equivalent.values` block. The Writer promoted it; the Checker re-verified on cycle-2. Without the writer/checker split, the number would have been citable but not structured-provenanced.

What you get. **102 entries:** 94 PRIMARY (80 flavor + 14 collider RS) + 8 SECONDARY. **5 Opus sign-off rounds, 102/102 APPROVE, 0 RETURN-TO-WA, 0 ESCALATE-TO-PI. Fact-check: 100 VERIFIED / 2 PARTIAL (both metadata-only, documented) / 0 MISMATCH / 0 FAILED.** Three arbitrations (L001; B001+B003; B021+B023) codified a CHK-1 carve-out separating measured observables from theory-normalization scales and dataset descriptors; downstream batches applied it uniformly without re-litigation. Every source snapshot is tracked in git with `sha256`; `git checkout <tag>` rebuilds the catalog from scratch with full provenance.

Pointers. Playbook `flavor_catalog/AGENTIC_WORKFLOW.md`; catalog PDF (v0.4, 179 pp.) `flavor_catalog/catalog_master.pdf`; per-round verdicts `signoff/round_NNN_index.md`; fact-check tables `audits/factcheck_*.md`.